EX.NO:3

Exploring the different types of prompts- Straightforward Prompts - Tabular Format Prompting - Missing Word Prompting - Preceding Question Prompting.

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Straight Forward Prompts:

1)What is entropy in thermodynamics?

Entropy in thermodynamics is a measure of disorder or randomness in a system.

More precisely, it tells how much energy in a system is unavailable to do useful work.

Simple Definition

Entropy (S) = measure of disorder

Higher entropy → more disorder

Lower entropy → more order

Physical Meaning

When systems naturally move from order → disorder, entropy increases.

It tells how spread out energy is.

Example:

Ice (solid, ordered) → water (less ordered) → steam (very disordered)

Entropy increases in this direction.

Thermodynamic Definition

Entropy change is:

$$\Delta S = \frac{Q_{rev}}{T}$$

Where:

= change in entropy

= heat absorbed/released in a reversible process

= temperature in Kelvin

Second Law of Thermodynamics

This law is based on entropy:

❗ In an isolated system, entropy never decreases.

Entropy either stays constant (ideal case) or increases (real processes).

Examples

1. Melting ice

Ice → water

Order breaks → entropy increases.

2. Mixing gases

Two gases mix automatically → more randomness → entropy increases.

3. Heat flow

Heat flows from hot → cold automatically.

This increases entropy because energy becomes more spread out.

If you want, I can also explain:

Entropy in simple school-level terms

Entropy vs Enthalpy

Entropy in information theory (different concept)

What language is used in unity?

Unity uses C# (C-Sharp) as its primary and officially supported programming language.

2) Why C# in Unity?

✔ Easy to learn (clean syntax, beginner-friendly)

✔ Powerful OOP features (classes, inheritance, polymorphism)

✔ Fast and efficient for game logic

✔ Huge community support

✔ Works with the Unity Engine API

Does Unity support other languages?

Historically: Unity used JavaScript-like language called UnityScript and Boo (Python-like).

Now: Only C# is supported for scripting.

If you want, I can give you:

A beginner roadmap to learn C# for Unity,

Simple example scripts,

Or a daily practice plan.

Tabular Prompts:

1)Compare unity and unreal in terms of ease, language used, community support, optimization, fast building.

Dimension	Unity	Unreal Engine
Ease of Use	• Widely regarded as more beginner-friendly: simpler UI, less steep learning curve. (rocketbrush.com) • Especially good for 2D, mobile, indie development. (apptunix) • Many tutorials, templates, asset store items to accelerate getting started.	• More complex in many respects: advanced rendering, deeper system access, steeper learning curve. (incredibuild) • Great power for high-end 3D/AAA but may require more expertise to fully leverage.
Language Used	• Primary scripting language: **C#**. (Wikipedia) • Editor and runtime written in C++ under the hood, but you as developer mostly write C#.	• Core engine is written in **C++**. (Wikipedia) • You also have visual scripting (Blueprints) for many tasks, useful for non-programmers.
Community Support	• Large community, many indie developers, lots of assets/plugins in the marketplace. • Because it's used for mobile/2D/indie, a lot of accessible content and help.	• Strong community especially in high-end game dev, AAA, film/VR. • Because of its power, many advanced resources, but maybe fewer "easy beginner" tutorials than Unity.
Optimization (performance / scale)	• Good for many platforms including mobile, but when you push to ultra-realistic AAA visuals, may need more work/optimization. (apptunix) • A recent academic study found that on physics/mesh workloads Unity	• Very strong for high-fidelity visuals, large open worlds, VR/AR, cinematic experiences. (Hackr.io) • Because you have deeper access (C++), you can optimize very well—but that

Dimension	Unity	Unreal Engine
	consumed significantly less energy in some cases vs Unreal. (arXiv)	also means the complexity is higher.
Fast Building / Iteration Speed	• Generally fast to get prototypes up and running, especially in 2D or simpler 3D projects. • Asset store helps speed up development. • Good for rapid iteration, smaller teams.	• You can build very powerful scenes, but setting up the infrastructure (lighting, shaders, large world streaming) may take longer. • For massive projects the build times / complexity may grow.

2) Compare IIT and NIT on placements, entrance, difficulty and facilities

Aspect	IIT	NIT
Entrance Exam	JEE Advanced	JEE Main
Entry Difficulty	Very high	High
Academic Difficulty	Very tough	Moderate–high
Placements	Very high packages	Good packages
Facilities	Excellent, research-focused	Good infrastructure
Brand Value	Very high (global)	High (India)

Missing Word Prompts:

1)“The sun rises in the ______.”

The missing word is East

2)The train ___ at 6 PM.”

a) leave

b) leaves

c) leaving

Ans: b)leaves

3) In photosynthesis, plants absorb sunlight to produce ____.

The missing word is glucose.

Preceding Question Prompting:

1) First, answer this question: What is the main product of photosynthesis?
   Now use that answer to complete the sentence:
   "In photosynthesis, plants absorb sunlight to produce ____."

Glucose

2) First, identify the gas released during photosynthesis.
   Then complete the sentence using that answer:
   "During photosynthesis, plants release ____ into the atmosphere."

oxygen